

1. A progressive wave $y(x, t) = 0.2 \sin(30t - 4.0x)$ where x is in cm and t is in s. Calculate:
 - (a) its maximum velocity and
 - (b) its maximum acceleration.

$$[v_{max} = \pm 6 \text{ cm s}^{-1}; a_{max} = 180 \text{ cm s}^{-2}]$$

2. A wave travelling in the positive x - direction has a frequency of 20 Hz as shown in **FIGURE 2.1**.

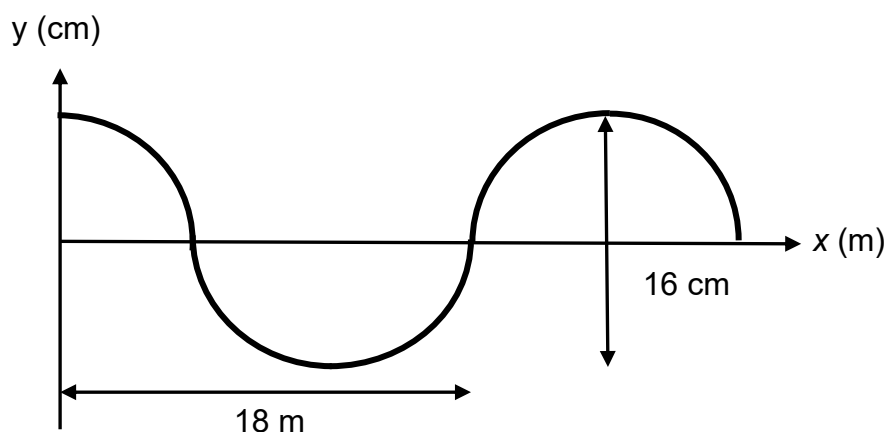


FIGURE 2.1

Determine the:

- (a) Amplitude
 - (b) Wavelength
 - (c) Period
 - (d) Speed of the wave.
- $$[A = 8 \text{ cm}; \lambda = 24 \text{ m}; T = 0.05 \text{ s}; v = 480 \text{ m s}^{-1}]$$
3. A harmonic wave is travelling along a rope. It completes 50 vibrations within 30 s. The generated wave profile travels 300 m along the rope within 45 s. What is its wavelength?

$$[\lambda = 11.12 \text{ m}]$$
 4. A wave propagation is represented by $y = 4 \sin(20\pi t - \frac{\pi}{6}x)$ where x is in cm and t is in s. Calculate the
 - (a) Wavelength
 - (b) Frequency
 - (c) Velocity

$$[\lambda = 12 \text{ cm}; f = 10 \text{ Hz}; v = 120 \text{ cm s}^{-1}]$$

5. The wave equation of a sinusoidal wave travelling is given as

$$y = 0.15 \sin (\pi t - 2 \pi x)$$

where y is in m and t in s. Sketch the graph of

- (a) at $t = 0$ for $0 \leq x \leq 2\pi$
- (b) at $x = 0$ when $0 \leq t \leq 2T$

[DIY]

6. **FIGURE 2.2** and **FIGURE 2.3** show the representation of the same wave.

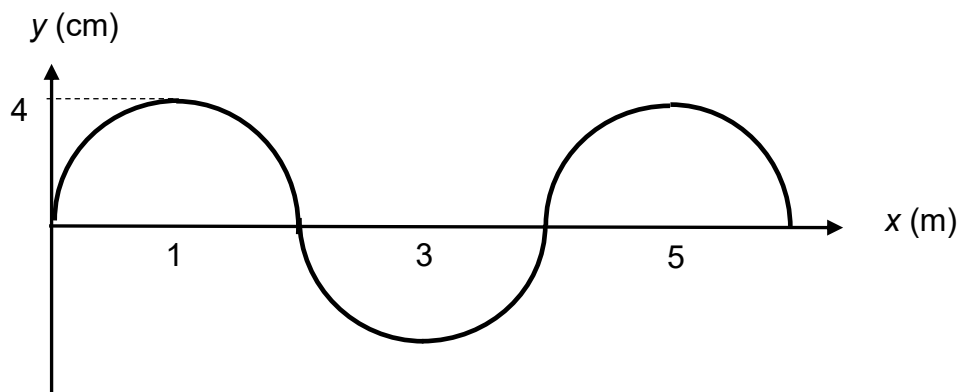


FIGURE 2.2

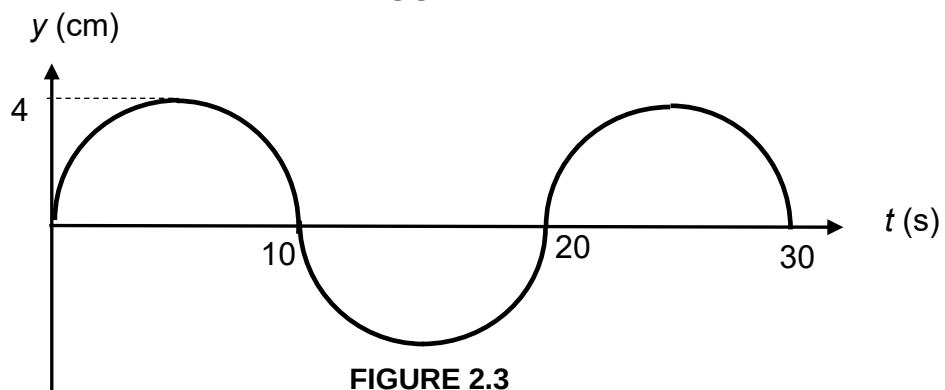


FIGURE 2.3

From the graph/s determine the:

- (a) Amplitude
- (b) Wavelength
- (c) Wave number
- (d) Period
- (e) Frequency
- (f) Angular frequency
- (g) Velocity.

$$[A = 4 \text{ cm}; \lambda = 4 \text{ m}; k = 0.5 \pi \text{ m}^{-1}; T = 20 \text{ s};$$

$$f = 0.05 \text{ Hz}; \omega = 0.3142 \text{ rad s}^{-1}; v = 0.2 \text{ m s}^{-1}]$$

7. A sinusoidal wave travelling in the $+x$ direction has an amplitude of 10 cm, wavelength 40 cm and frequency of 8 Hz. Assume that the particle $x = 0$; at $t = 0$ s its displacement, $y = 0$. Determine the equation to represent the motion.

$$[y = 10 \sin (16 \pi t - 0.05 \pi x)]$$

8. A progressive wave propagates in the $+x$ direction with a velocity of 10 m s^{-1} . The amplitude and wavelength are 12 mm and 40 cm respectively.

- (a) Determine the wave equation.
 (b) Calculate the vibrational velocity of a particle at position $x = 0.5 \text{ cm}$ at the time $t = 4 \text{ s}$.

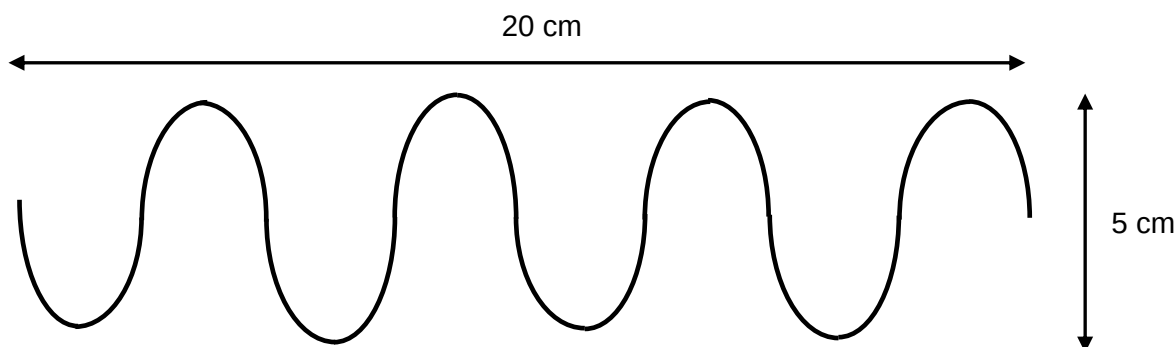
$$[y = (12 \times 10^{-3}) \sin (50 \pi t - 5 \pi x); v = 1.196 \times 10^{-2} \text{ m s}^{-1}]$$

9. A progressive wave propagates in negative $-x$ direction with velocity of 6 m s^{-1} . The amplitude and wavelength are 12 cm and 35 cm respectively. Determine

- (a) The wave equation
 (b) The vibrational speed of a particle at $x = 10 \text{ cm}$ at time $t = 0.5 \text{ s}$
 (c) The direction of the wave.

$$[y = 0.12 \sin (107.7t + 17.95x); v = \pm 2.80 \text{ m s}^{-1}; \text{DIY}]$$

10. The progressive wave below is travelling to the right at a speed of 22 m s^{-1} .



- (a) Determine the
 (i) wave number
 (ii) frequency of the wave
 (b) Write the wave equation

$$[k = 40 \pi \text{ m}^{-1} \quad f = 440 \text{ Hz}; y = 0.025 \sin (880 \pi t - 40 \pi x)]$$